

# Package ‘spca’

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**Type** Package

**Title** Least Squares Sparse Principal Components Analysis

**Version** 1.1.4

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**Description** Implements least-squares sparse principal component analysis with cardinality constraints. The package has an efficient C++ backend and provides functions for fitting, summarizing, comparing, and visualizing sparse principal component models. The approach follows Merola (2015) <doi:10.1111/anzs.12128> and Merola and Chen (2019) <doi:10.1016/j.jmva.2019.04.001>.

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**URL** <https://github.com/merolagio/spca>

**BugReports** <https://github.com/merolagio/spca/issues>

**License** AGPL-3

**Encoding** UTF-8

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|              |                                                           |
|--------------|-----------------------------------------------------------|
| spca-package | <i>Least Squares Sparse Principal Components Analysis</i> |
|--------------|-----------------------------------------------------------|

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Description

The package provides functions to compute LS-SPCA solutions, in which sparsity is imposed on Pearson PCA’s least-squares reconstruction objective.

Details

LS-SPCA differs from SPCA methods that compute sparse PCs by maximizing variance. Details are provided in the references below and in the extended vignette.

This release accompanies the related article and supports reproduction of the results reported therein.

Computation relies on efficient C++ routines and includes multiple options for variable selection and sparse weight estimation.

Fitting functions

- [spca\(\)](#) Computes LS-SPCA solutions from a data or covariance/correlation matrix. It returns a [spca\\_object](#) of class `spca`.
- [pca\(\)](#) Computes PCA solutions from a data or covariance/correlation matrix. It returns a [spca\\_object](#) inheriting from classes `spca_pca` and `spca`.

Methods

- `print()`, `summary()`, and `plot()` inspect and display spca objects.
- `change_sign()` changes the signs of selected components and their related object elements.
- `show_weights()` prints or returns the nonzero weights or contributions for selected components.
- `aggregate_by_group()` aggregates weights or contributions according to a grouping vector.
- `scree_plot()` and `mp_qqplot()` provide diagnostic plots for objects returned by `pca()`.

### Utilities

- `is.spca()` Verifies whether an object is a proper spca object.
- `is.pca()` Verifies whether an object is a proper pca object.
- `compare_spca()` Compares two or more LS-SPCA solutions numerically and visually.
- `new_spca()` Creates an spca object from a set of weights.

The former interfaces `change_weights_sign_spca()`, `change_loadings_sign_spca()`, `spca_screeplot()`, and `wachter_qqplot()` are retained for backward compatibility and issue deprecation warnings. Objects created by previous package versions with `loadings` and `loadings_list` elements remain supported.

### Author(s)

**Maintainer:** Giovanni Maria Merola <merolagio@gmail.com>

### References

Merola, G. M. (2015). Least Squares Sparse Principal Component Analysis: a Backward Elimination approach to attain large weights. *Australia & New Zealand Journal of Statistics*, 57, 391–429. doi:10.1111/anzs.12128

Merola, G. M. and Chen, G. (2019). Projection sparse principal component analysis: An efficient least squares method. *Journal of Multivariate Analysis*, 173, 366–382. doi:10.1016/j.jmva.2019.04.001

### See Also

Useful links:

- <https://github.com/merolagio/spca>
- Report bugs at <https://github.com/merolagio/spca/issues>

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|                    |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| aggregate_by_group | <i>Aggregate Weights or Contributions of a pca or spca Object by Group</i> |
|--------------------|----------------------------------------------------------------------------|

---

## Description

Aggregate component weights or contributions according to a grouping variable.

## Usage

```
aggregate_by_group(
  object,
  variable_groups,
  cols = NULL,
  only_nonzero = TRUE,
  contributions = TRUE,
  digits = ifelse(contributions, 1, 3),
  print_table = TRUE,
  return_table = FALSE,
  spca_obj = NULL
)
```

```
## S3 method for class 'spca'
aggregate_by_group(
  object,
  variable_groups,
  cols = NULL,
  only_nonzero = TRUE,
  contributions = TRUE,
  digits = ifelse(contributions, 1, 3),
  print_table = TRUE,
  return_table = FALSE,
  spca_obj = NULL
)
```

```
## S3 method for class 'pca'
aggregate_by_group(object, ...)
```

## Arguments

|                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| object          | An object of class pca or spca.                                                                                                                         |
| variable_groups | A vector or factor with one group label per variable.                                                                                                   |
| cols            | An integer vector selecting components. If a single positive integer is supplied, components 1:cols are included. If NULL, all components are included. |
| only_nonzero    | A logical value indicating whether to omit variable_groups whose values are zero in every selected component.                                           |

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| contributions | A logical value. If TRUE, aggregate percentage contributions; otherwise, aggregate weights.                                       |
| digits        | Number of digits used in printed output.                                                                                          |
| print_table   | A logical value indicating whether to print the table.                                                                            |
| return_table  | A logical value indicating whether to return the table visibly.                                                                   |
| spca_obj      | Deprecated alias for object (default NULL). Supply only one of object and spca_obj. Using the old argument name issues a warning. |
| ...           | Additional arguments passed to aggregate_by_group.spca.                                                                           |

### Methods (by class)

- aggregate\_by\_group(pca): Aggregate a pca object by group.

### See Also

Other spca: [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

Other pca: [change\\_sign\(\)](#), [is.pca\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [print.pca\(\)](#), [scree\\_plot\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

---

change\_sign

---

*Change Component Signs in a pca or spca Object*


---

### Description

Change the signs of selected Weights in a fitted object.

### Usage

```
change_sign(object, index_to_change, spca_obj = NULL)
```

```
## S3 method for class 'spca'
```

```
change_sign(object, index_to_change, spca_obj = NULL)
```

```
## S3 method for class 'pca'
```

```
change_sign(object, ...)
```

### Arguments

|                 |                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| object          | An object of class pca or spca.                                                                                                   |
| index_to_change | An integer vector of component indices whose signs should be changed.                                                             |
| spca_obj        | Deprecated alias for object (default NULL). Supply only one of object and spca_obj. Using the old argument name issues a warning. |
| ...             | Additional arguments passed to change_sign.spca.                                                                                  |

**Value**

The modified object, preserving its classes.

**Methods (by class)**

- `change_sign(pca)`: Change component signs in a `pca` object.

**See Also**

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

Other `pca`: [aggregate\\_by\\_group\(\)](#), [is.pca\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [print.pca\(\)](#), [scree\\_plot\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

**Examples**

```
data(holzinger)
ho_cspca = spca(holzinger, n_comps = 4)
show_correlations(ho_cspca)
# In applications we would change only the fourth set of weights
ho_changed = change_sign(ho_cspca, index_to_change = c(2, 4))
is.spca(ho_changed)
show_weights(ho_cspca, cols = 2)
show_weights(ho_changed, cols = 2)
show_correlations(ho_changed)
```

---

compare\_spca

*Compare Two or More spca Solutions*

---

**Description**

Plots weights and print summary statistics for two or more `spca` objects side by side. For the meaning of each summary statistic, see [summary.spca](#). Tables and plots can optionally be returned.

**Usage**

```
compare_spca(
  obj_list,
  n_comps = NULL,
  contributions = TRUE,
  only_nonzero = TRUE,
  variable_groups = NULL,
  plot_weights = TRUE,
  plot_type = c("bars", "points"),
  methods_names = NULL,
  x_axis_var_names = FALSE,
```

```

col_grouplines = "black",
color_scale = c("ggplot", "cbb", "printsafe", "bw"),
col_short_names = TRUE,
print_tables = TRUE,
print_weights = FALSE,
print_summary = TRUE,
show_plot = TRUE,
return_tables = FALSE,
return_plot = FALSE
)

```

### Arguments

|                               |                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>obj_list</code>         | A list of two or more <code>sPCA</code> objects.                                                                                                                                                                    |
| <code>n_comps</code>          | An integer scalar or <code>NULL</code> (default <code>NULL</code> ). Number of components to compare. If <code>NULL</code> , the minimum number of available components across objects is used.                     |
| <code>contributions</code>    | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , compare percentage contributions; otherwise, compare weights.                                                                                  |
| <code>only_nonzero</code>     | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , only variables with at least one nonzero weight or contribution are plotted or printed.                                                        |
| <code>variable_groups</code>  | Optional variable grouping (default <code>NULL</code> ). Can be a list of indices, a vector, or a factor with one entry per variable. Used to draw vertical group-separating lines in the weights plot.             |
| <code>plot_weights</code>     | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , plot the weights or contributions.                                                                                                             |
| <code>plot_type</code>        | A character vector (default first element <code>"bars"</code> ). Values starting with <code>"b"</code> use bars; values starting with <code>"p"</code> use points. Other values default to bars.                    |
| <code>methods_names</code>    | An optional character vector (default <code>NULL</code> ) with one label per object. If <code>NULL</code> , labels are <code>M1</code> , ..., <code>Mk</code> .                                                     |
| <code>x_axis_var_names</code> | A logical value (default <code>FALSE</code> ). If <code>TRUE</code> , show variable names on the x axis of the weights plot.                                                                                        |
| <code>col_grouplines</code>   | A character scalar (default <code>"black"</code> ). Color of the vertical group lines.                                                                                                                              |
| <code>color_scale</code>      | A character vector (default first element <code>"ggplot"</code> ). Color palette for bar plots. Accepted values are <code>"ggplot"</code> , <code>"cbb"</code> , <code>"printsafe"</code> , and <code>"bw"</code> . |
| <code>col_short_names</code>  | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , use short component names such as <code>C1.M1</code> ; otherwise, use names such as <code>C1.object_name</code> .                              |
| <code>print_tables</code>     | A logical value (default <code>TRUE</code> ). If <code>FALSE</code> , suppress all table printing. Takes priority over <code>print_weights</code> and <code>print_summary</code> .                                  |
| <code>print_weights</code>    | A logical value (default <code>FALSE</code> ). If <code>TRUE</code> , print the weights or contributions table.                                                                                                     |
| <code>print_summary</code>    | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , print the summary statistics table when <code>print_tables = TRUE</code> .                                                                     |
| <code>show_plot</code>        | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , show the weights or contributions plot.                                                                                                        |

**return\_tables** A logical value (default FALSE). If TRUE, return the weights or contributions matrix and the raw summary matrix.

**return\_plot** A logical value (default FALSE). If TRUE, return the weights or contributions plot.

### Value

Invisibly returns NULL by default. If `return_tables = TRUE`, returns a list containing the comparison matrix and summary matrix. If `return_plot = TRUE`, the returned object also includes the weights or contributions plot.

### See Also

Other spca: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

### Examples

```
data(holzinger)
ho_uspca = spca(holzinger, n_comps = 4, method = "u")
ho_cspca = spca(holzinger, n_comps = 4, method = "c")
compare_spca(list(ho_uspca, ho_cspca))
```

---

holzinger

*Holzinger–Swineford Student Ability Data*

---

### Description

This dataset is based on the classic Holzinger and Swineford (1939) Student Ability dataset.

We use the version distributed with the `psychTools` package. For comparability with previous analyses, we select 12 items and only students from the Grant–White School (see also Ferrara, Martella, and Vichi, 2019).

### Usage

```
holzinger
```

### Format

**holzinger** A numeric data frame with 145 rows and 12 variables. The variables are:

**visual** Visual perception test (SPL).

**cubes** Cubes test (SPL).

**flags** Lozenges test (SPL).

**paragraph** Paragraph comprehension test (VBL).

**sentence** Sentence completion test (VBL).

**wordm** Word meaning test (VBL).  
**addition** Addition test (SPD).  
**counting** Counting groups of dots test (SPD).  
**straight** Straight and curved capitals test (SPD).  
**deduct** Deduction test (MTH).  
**numeric** Numerical puzzles test (MTH).  
**series** Series completion test (MTH).

### Details

The 12 items correspond to four ability scales: spatial (SPL), verbal (VBL), speed (SPD), and mathematical (MTH). The data provided with this package are scaled to mean zero and unit variance. The scales are available as a factor called `holzinger_scales`

### References

Holzinger, K. J., and Swineford, F. (1939). *A study in factor analysis: The stability of a bi-factor solution*. Supplementary Educational Monographs, No. 48.

Ferrara, C., Martella, F., and Vichi, M. (2019). Probabilistic disjoint principal component analysis. *Multivariate Behavioral Research*, 54(1), 47–61.

---

|                  |                                                   |
|------------------|---------------------------------------------------|
| holzinger_scales | <i>Holzinger–Swineford Student Ability Scales</i> |
|------------------|---------------------------------------------------|

---

### Description

Holzinger–Swineford Student Ability Scales

### Usage

```
holzinger_scales
```

### Format

**holzinger\_scales** A factor listing the 4 scales: SPL, VBL, SPD and MTH, for each variable.

---

is.pca

*Test for PCA Objects*


---

### Description

Check whether an object has class `pca` and contains the core elements produced by `pca()`.

### Usage

```
is.pca(x)
```

### Arguments

`x`                      An object to test.

### Details

Performs a lightweight check of the class and the presence of core elements. It does not validate their values or dimensions. Scores and the number of observations are not required because they may be unavailable for covariance-matrix input.

### Value

A logical value: TRUE if the required class and core elements are present, and FALSE otherwise.

### See Also

Other `pca`: [aggregate\\_by\\_group\(\)](#), [change\\_sign\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [print.pca\(\)](#), [scree\\_plot\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

### Examples

```
data(holzinger)
ho_pca = pca(holzinger, n_comps = 2, screeplot = FALSE, qq_plot = FALSE)
is.pca(ho_pca)
```

---

is.spca

*Test for SPCA Objects*


---

### Description

Check whether an object has class `spca` and contains the core elements required by the package.

### Usage

```
is.spca(x)
```

**Arguments**

`x`                      An object to test.

**Details**

The function checks for class `spca` and for the presence of the core elements used by the package, including weights, contributions, explained-variance summaries, component counts, cardinalities, weight lists, and active indices. It performs a lightweight structural check; use `validate_spca()` for a more detailed internal validation.

**Value**

A logical value. Returns TRUE if `x` has class `spca` and contains the required core elements, and FALSE otherwise.

**See Also**

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

**Examples**

```
data(holzinger)
ho_cspca = spca(holzinger, n_comps = 2)
is.spca(ho_cspca)
```

---

mp\_qqplot

---

*PCA Diagnostic qqplot*


---

**Description**

Produce a qq-plot comparing the eigenvalues of a fitted PCA with Marchenko–Pastur theoretical quantiles.

**Usage**

```
mp_qqplot(
  pca_fit,
  n_vars = NULL,
  n_obs = NULL,
  gamma = NULL,
  cor = TRUE,
  common_var = NULL,
  n_plot = NULL,
  n_fitline = NULL,
  addtitle = TRUE,
```

```

    show_plot = TRUE,
    return_plot = FALSE
)

## S3 method for class 'pca'
mp_qqplot(
  pca_fit,
  n_vars = NULL,
  n_obs = NULL,
  gamma = NULL,
  cor = TRUE,
  common_var = NULL,
  n_plot = NULL,
  n_fitline = NULL,
  addtitle = TRUE,
  show_plot = TRUE,
  return_plot = FALSE
)

```

### Arguments

|                          |                                                                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pca_fit</code>     | An object of class "pca", as returned by <code>pca()</code> .                                                                                                                                                                                                    |
| <code>n_vars</code>      | An integer scalar or NULL (default NULL). Number of variables. If NULL, obtain it from the number of rows of <code>pca_fit\$weights</code> .                                                                                                                     |
| <code>n_obs</code>       | An integer scalar or NULL (default NULL). Number of observations. If NULL, obtain it from <code>pca_fit\$n_obs</code> .                                                                                                                                          |
| <code>gamma</code>       | A positive numeric scalar or NULL (default NULL). Ratio of observations to variables. If NULL, use <code>n_obs / n_vars</code> .                                                                                                                                 |
| <code>cor</code>         | A logical scalar (default TRUE). If TRUE, use the correlation-matrix reference: set <code>common_var = 1</code> and scale the theoretical quantiles to sum to <code>n_vars</code> . If FALSE, retain the covariance scale specified by <code>common_var</code> . |
| <code>common_var</code>  | A positive numeric scalar or NULL (default NULL). Assumed common population variance for the Marchenko–Pastur quantiles. Required when <code>cor = FALSE</code> ; set to 1 when <code>cor = TRUE</code> .                                                        |
| <code>n_plot</code>      | An integer scalar or NULL (default NULL). Number of leading eigenvalues. If NULL, use all stored or supplied eigenvalues.                                                                                                                                        |
| <code>n_fitline</code>   | An integer scalar or NULL (default NULL: no line). Zero also omits the line. If positive, fit a line using the last <code>n_fitline</code> points. If negative, exclude the largest <code>abs(n_fitline)</code> points.                                          |
| <code>addtitle</code>    | A logical scalar (default TRUE). Add a title.                                                                                                                                                                                                                    |
| <code>show_plot</code>   | A logical scalar (default TRUE). Print the plot.                                                                                                                                                                                                                 |
| <code>return_plot</code> | A logical scalar (default FALSE). Return the plot.                                                                                                                                                                                                               |

### Details

This diagnostic relies on Marchenko–Pastur theory for the null eigenvalue distribution of a (possibly high-dimensional) sample covariance matrix, and is therefore only meaningful for objects of class

"pca" (i.e. objects returned by `pca()`). It does not apply to sparse fits returned by `spca()` (class "spca"), for which the reference distribution does not hold; no method is defined for that class, so calling it on such an object raises the standard "no applicable method" error rather than producing a plot.

The Marchenko–Pastur distribution depends on the ratio of variables to observations. The `gamma` argument uses its reciprocal,  $n_{\text{obs}} / n_{\text{vars}}$ , as required by `RMTstat::qmp`. Both counts must be available, even when `gamma` is supplied. The covariance reference assumes independent variables with a common population variance. With `cor = TRUE`, the theoretical quantiles are scaled to sum to  $n_{\text{vars}}$ . A warning is issued if the stored eigenvalues do not sum to  $n_{\text{vars}}$  within a tolerance of  $1e-8 * n_{\text{vars}}$ . This check assumes that all nonzero eigenvalues are stored; it does not establish that the original matrix is a correlation matrix. With `cor = FALSE`, the quantiles retain the scale set by `common_var`. A fitted line must use at least two of the plotted points.

### Value

If `return_plot = TRUE`, a `ggplot` object; otherwise `NULL` invisibly.

### See Also

Other `pca`: `aggregate_by_group()`, `change_sign()`, `is.pca()`, `pca()`, `print.pca()`, `scree_plot()`, `show_correlations()`, `show_weights()`, `spca_screeplot()`, `wachter_qqplot()`

### Examples

```
data(holzinger)
ho_pca = pca(holzinger)
# from the screeplot we may choose to retain 4 components
# produce a Wachter qqplot fitting a line to all but the largest
# 4 eigenvalues.
# Since pca was fitted using the data matrix, the fit carries the number
# of observations. Otherwise we would need to pass `n_obs` .
mp_qqplot(ho_pca, n_fitline = -4)
# The qq-plot indicates that the 4th eigenvalue is compatible
# with that of a random matrix.
```

---

new\_sPCA

*Construct an SPCA Object from a Set of Weights*

---

### Description

Build an object of class `spca` from a weights matrix and either a covariance or correlation matrix, a data matrix, or both.

### Usage

```
new_sPCA(A, S = NULL, X = NULL, method_name = NULL)
```

**Arguments**

|             |                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A           | A numeric matrix of weights.                                                                                                                                       |
| S           | A numeric covariance or correlation matrix (default NULL). If NULL, X is used to estimate the covariance matrix.                                                   |
| X           | A numeric data matrix or data frame (default NULL). Used to compute S when S = NULL, and to compute scores when supplied. At least one of S or X must be provided. |
| method_name | A character scalar or NULL (default NULL). Name of the method used to compute the weights.                                                                         |

**Value**

An spca object.

**See Also**

Other spca: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

**Examples**

```
set.seed(1)
A = round(matrix(runif(24, -1, 1), 12))
A[abs(A) < 0.4] = 0 #no need to scale to unit norm
data(holzinger)
spca_new = new_spca(A, X = holzinger)
is.spca(spca_new)
summary(spca_new)
```

---

pca

*Computes Principal Components*

---

**Description**

Compute a principal component analysis (PCA) and return the result as an spca object, so that it can be used with spca methods.

**Usage**

```
pca(
  M,
  n_comps = NULL,
  center_data = FALSE,
  scale_data = FALSE,
  fat_matrix = NULL,
```

```

    screeplot = TRUE,
    qq_plot = FALSE,
    n_obs = NULL,
    neigen_toplot = NULL,
    cor = TRUE,
    common_var = NULL,
    pm = FALSE,
    eps_pm = 1e-04,
    maxiter_pm = 1000
)

```

### Arguments

|               |                                                                                                                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M             | A data matrix, correlation matrix, or covariance matrix.                                                                                                                                                                                                                                                             |
| n_comps       | An integer scalar or NULL (default NULL). Number of components to retain. If NULL, all components are retained up to the maximum allowed by the selected backend.                                                                                                                                                    |
| center_data   | A logical value (default FALSE). If TRUE, center variables to zero mean. If M is detected as a data matrix and any column mean is nonzero, centering is performed automatically.                                                                                                                                     |
| scale_data    | A logical value (default FALSE). If TRUE, scale variables.                                                                                                                                                                                                                                                           |
| fat_matrix    | A logical value or NULL (default NULL). If NULL, the backend is selected automatically: data matrices with $n < p$ use the fat backend and all other inputs use the tall backend. If TRUE, request the fat backend. If FALSE, use the tall backend. Covariance and correlation matrices always use the tall backend. |
| screeplot     | A logical value (default TRUE). If TRUE, produce a scree plot.                                                                                                                                                                                                                                                       |
| qq_plot       | A logical value (default FALSE). If TRUE, produce a Wachter qq-plot with <a href="#">mp_qqplot</a> .                                                                                                                                                                                                                 |
| n_obs         | An integer scalar or NULL (default NULL). Number of rows in the original data set. Required when qq_plot = TRUE and M is a covariance or correlation matrix. If not available, the Wachter qq-plot cannot be produced.                                                                                               |
| neigen_toplot | An integer scalar or NULL (default NULL). Number of eigenvalues to show in diagnostic plots. If NULL, all available eigenvalues are shown.                                                                                                                                                                           |
| cor           | A logical value (default TRUE); the diagnostic plot uses common_var for the Marchenko–Pastur quantiles.                                                                                                                                                                                                              |
| common_var    | A positive numeric scalar or NULL (default NULL). Common population variance assumed for the Wachter qq-plot. Required when qq_plot = TRUE and cor = FALSE. When cor = TRUE, the reference variance is set to 1.                                                                                                     |
| pm            | A logical value (default FALSE). If TRUE, compute the requested eigenpairs by power method and rank-one deflation.                                                                                                                                                                                                   |
| eps_pm        | A positive numeric scalar (default 1e-4). Convergence tolerance for the power method.                                                                                                                                                                                                                                |
| maxiter_pm    | A positive integer scalar (default 1000). Maximum number of power-method iterations.                                                                                                                                                                                                                                 |

## Details

`n_comps` controls how many components are retained in the returned object. The tall backend computes PCA from the covariance or correlation matrix, so passing this saves its computation from the data matrix. If a covariance or correlation matrix is passed, the scores cannot be computed and are returned as NULL. The fat backend computes PCA in row space and converts the retained eigenvectors back to variable weights.

## Value

An `pca_object` which is the same as an `spca_object` with an additional eigenvalues vector containing the eigenvalues up to the rank used by the selected backend and `n_obs` stores the number of observations, if a data matrix is passed, or NULL.

## See Also

Other `pca`: `aggregate_by_group()`, `change_sign()`, `is.pca()`, `mp_qqplot()`, `print.pca()`, `scree_plot()`, `show_correlations()`, `show_weights()`, `spca_screeplot()`, `wachter_qqplot()`

## Examples

```
data(holzinger)
ho_pca = pca(holzinger, n_comps = 4, screeplot = TRUE,
             n_obs = 144, qq_plot = TRUE)
summary(ho_pca)
```

---

plot.pca

*Plot the Weights or Contributions of a pca or spca Object*

---

## Description

Plot the sparse weights, or the corresponding percentage contributions, from a fitted object. The plot can be shown as a bar plot, circular bar plot, or heatmap.

## Usage

```
## S3 method for class 'pca'
plot(x, ...)

## S3 method for class 'spca'
plot(
  x,
  n_plot = NULL,
  plot_type = c("bars", "circular", "heatmap"),
  contributions = TRUE,
  only_nonzero = TRUE,
  pc_weights = NULL,
  variable_groups = NULL,
```

```

    plot_title = NULL,
    return_plot = FALSE,
    show_plot = TRUE,
    controls = list(color_scale = c("ggplot", "cbb", "printsafe", "bw"), variable_names =
      c("none", "auto"), legend_position = c("none", "bottom", "right", "top", "left"),
      grid_type = c("horizontal", "full", "none"), facet_labels = NULL, legend_title =
      NULL, x_axis_lab = "variables", adjust_labels_circ = NULL, flip_heatmap = FALSE,
      heatmap_color_range = c("values", "unit")),
    ...
  )

```

## Arguments

|                              |                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>               | An object of class <code>spca</code> or <code>pca</code> .                                                                                                                                                                                                                                                                                                                            |
| <code>...</code>             | Further arguments. These are currently unused and trigger an error if supplied.                                                                                                                                                                                                                                                                                                       |
| <code>n_plot</code>          | An integer scalar or <code>NULL</code> (default <code>NULL</code> ). Number of components to plot. If <code>NULL</code> , all components in <code>x</code> are plotted.                                                                                                                                                                                                               |
| <code>plot_type</code>       | A character vector (default first element <code>"bars"</code> ). Plot type. Accepted values are <code>"bars"</code> , <code>"circular"</code> , and <code>"heatmap"</code> . The first character is enough for matching.                                                                                                                                                              |
| <code>contributions</code>   | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , plot percentage contributions; otherwise, plot L2 unit weights.                                                                                                                                                                                                                                                  |
| <code>only_nonzero</code>    | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , plot only variables with at least one nonzero weight.                                                                                                                                                                                                                                                            |
| <code>pc_weights</code>      | A numeric matrix, data frame, or <code>NULL</code> (default <code>NULL</code> ). Optional PCA weights or contributions to plot together with the SPCA values for comparison.                                                                                                                                                                                                          |
| <code>variable_groups</code> | A vector, factor, or <code>NULL</code> (default <code>NULL</code> ). Optional grouping variable of length $p$ , where $p$ is the number of variables. If supplied, bars or tiles are colored by group instead of by component.                                                                                                                                                        |
| <code>plot_title</code>      | A character scalar or <code>NULL</code> (default <code>NULL</code> ). Optional plot title.                                                                                                                                                                                                                                                                                            |
| <code>return_plot</code>     | A logical value (default <code>FALSE</code> ). If <code>TRUE</code> , return the <code>ggplot2</code> object.                                                                                                                                                                                                                                                                         |
| <code>show_plot</code>       | A logical value (default <code>TRUE</code> ). If <code>TRUE</code> , print the plot.                                                                                                                                                                                                                                                                                                  |
| <code>controls</code>        | A list of graphical controls (default described below). Supported entries are <code>color_scale</code> , <code>variable_names</code> , <code>legend_position</code> , <code>grid_type</code> , <code>facet_labels</code> , <code>legend_title</code> , <code>x_axis_lab</code> , <code>adjust_labels_circ</code> , <code>flip_heatmap</code> , and <code>heatmap_color_range</code> . |

## Details

If `pc_weights` is supplied, SPCA and PCA values are plotted side by side for comparison. Circular bar plots are not implemented for this comparison, so a standard bar plot is used instead. In this case all variables are plotted, regardless of `only_nonzero`.

For character arguments defined by a default vector of accepted values, the first element is the default and the first character of the supplied string is used for matching.

The entries in `controls` are:

- **color\_scale**: a character vector (default first element "ggplot"). Accepted values are "ggplot", "cbb", "printsafe", and "bw". "cbb" is colorblind-friendly with black, "printsafe" is colorblind- and printer-friendly, "bw" uses gray tones, and "ggplot" uses the default ggplot2 scale.
- **variable\_names**: a character vector or "auto" or "none" (default "none"). If "auto", row names of the weight matrix are used as variable names; if row names are missing, no variable names are printed. NULL is a deprecated alias for "auto" retained for backward compatibility; using it emits a warning to use "auto" instead. If set to "none", variable names are not shown. If a character vector of length  $p$  is supplied, it is used as the variable names.
- **legend\_position**: a character vector (default first element "none"). Accepted values are "none", "bottom", "right", "top", and "left".
- **grid\_type**: a character vector (default first element "horizontal"). Accepted values are "horizontal", "full", and "none".
- **facet\_labels**: a character vector or NULL (default NULL). Optional facet labels for components.
- **legend\_title**: a character scalar or NULL (default NULL). Optional legend title.
- **x\_axis\_lab**: a character scalar (default "variables"). Label for the x axis.
- **adjust\_labels\_circ**: a numeric vector or NULL (default NULL). Optional angular adjustments for circular plot labels.
- **flip\_heatmap**: a logical value (default FALSE). If TRUE, flip the heatmap axes.
- **heatmap\_color\_range**: a character vector (default first element "values"). Accepted values are "values" and "unit".

When variable groups are supplied, a legend is needed to identify the groups; if the legend is missing or suppressed, it is moved to the bottom. For circular plots, the legend is moved to the right unless it is suppressed.

## Value

If `return_plot = TRUE`, returns the ggplot2 object. Otherwise, returns NULL invisibly.

## Functions

- `plot(pca)`: A `pca` object is plotted identically to an `spca` object.

## References

The printsafe palette corresponds to OrRd from <https://colorbrewer2.org/>.

## See Also

Other `spca`: `aggregate_by_group()`, `change_loadings_sign_spca()`, `change_sign()`, `change_weights_sign_spca()`, `compare_spca()`, `is.spca()`, `new_spca()`, `print.pca()`, `show_contributions_spca()`, `show_correlations()`, `show_weights()`, `spca()`, `spca_object`, `summary.pca()`

Other `spca`: `aggregate_by_group()`, `change_loadings_sign_spca()`, `change_sign()`, `change_weights_sign_spca()`, `compare_spca()`, `is.spca()`, `new_spca()`, `print.pca()`, `show_contributions_spca()`, `show_correlations()`, `show_weights()`, `spca()`, `spca_object`, `summary.pca()`

**Examples**

```

data(holzinger)
ho_cspca = spca(holzinger, n_comps = 4)
ho_plot = plot(ho_cspca, return_plot = TRUE)

# Change faceting and legend position.
ho_plot + ggplot2::facet_wrap(
  facets = ggplot2::vars(component),
  ncol = 4,
  nrow = 1
) + ggplot2::theme(legend.position = "right")

```

print.pca

*Print the Weights or Contributions of a pca or spca Object***Description**

Print sparse weights, or the corresponding percentage contributions, from a fitted object. By default, variables with only zero entries are omitted, and cumulative explained variance is shown at the bottom of the table.

**Usage**

```

## S3 method for class 'pca'
print(x, ...)

## S3 method for class 'spca'
print(
  x,
  cols = NULL,
  only_nonzero = TRUE,
  contributions = TRUE,
  digits = 3,
  thresh_card = 1e-07,
  return_table = FALSE,
  component_names = NULL,
  ...
)

```

**Arguments**

|      |                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x    | An object of class spca or pca.                                                                                                                                     |
| ...  | Further arguments. These are currently unused and trigger an error if supplied.                                                                                     |
| cols | An integer vector or NULL (default NULL). Components to print. If NULL, all components are printed. If a single integer is supplied, components 1:cols are printed. |

|                 |                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| only_nonzero    | A logical value (default TRUE). If TRUE, print only variables with at least one weight or contribution whose absolute value is greater than or equal to thresh_card. |
| contributions   | A logical value (default TRUE). If TRUE, print weights scaled to unit $L_1$ norm as percentage contributions; otherwise, print L2 unit weights.                      |
| digits          | An integer scalar (default 3). Number of decimal places used when printing weights. Contributions are printed as percentages with one decimal place.                 |
| thresh_card     | A numeric scalar (default $1e-07$ ). Values with absolute magnitude below this threshold are treated as zero in the printed table.                                   |
| return_table    | A logical value (default FALSE). If TRUE, return the formatted character matrix.                                                                                     |
| component_names | A character vector or NULL (default NULL). Optional component names. If NULL, existing column names are used when available; otherwise default names are assigned.   |

### Value

If return\_table = TRUE, returns the formatted character matrix. Otherwise, returns NULL invisibly.

### Functions

- print(pca): A pca object is printed identically to an spca object.

### See Also

Other spca: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

Other pca: [aggregate\\_by\\_group\(\)](#), [change\\_sign\(\)](#), [is.pca\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [scree\\_plot\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

### Examples

```
data(holzinger)
ho_cspca = spca(holzinger, n_comps = 4)
ho_cspca
print(ho_cspca, contributions = FALSE, digits = 4)
```

---

scree\_plot

*S3 Method for Plotting PCA Eigenvalues in a Screeplot*

---

### Description

Plot the leading eigenvalues of a fitted PCA against component order.

**Usage**

```

scree_plot(
  pca_fit,
  n_plot = NULL,
  ylab = "eigenvalues",
  addtitle = TRUE,
  show_plot = TRUE,
  return_plot = FALSE
)

## S3 method for class 'pca'
scree_plot(
  pca_fit,
  n_plot = NULL,
  ylab = "eigenvalues",
  addtitle = TRUE,
  show_plot = TRUE,
  return_plot = FALSE
)

```

**Arguments**

|                          |                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>pca_fit</code>     | An object of class "pca", as returned by <code>pca()</code> .                                                             |
| <code>n_plot</code>      | An integer scalar or NULL (default NULL). Number of leading eigenvalues. If NULL, use all stored or supplied eigenvalues. |
| <code>ylab</code>        | A character scalar (default "eigenvalues"). Y-axis label.                                                                 |
| <code>addtitle</code>    | A logical scalar (default TRUE). Add a title.                                                                             |
| <code>show_plot</code>   | A logical scalar (default TRUE). Print the plot.                                                                          |
| <code>return_plot</code> | A logical scalar (default FALSE). Return the plot.                                                                        |

**Details**

This diagnostic applies only to objects of class "pca" (i.e. objects returned by `pca()`). It does not apply to sparse fits returned by `spca()` (class "spca"); no method is defined for that class, so calling it on such an object raises the standard "no applicable method" error rather than producing a plot.

**Value**

If `return_plot = TRUE`, a ggplot object; otherwise NULL invisibly.

**See Also**

Other `pca`: `aggregate_by_group()`, `change_sign()`, `is.pca()`, `mp_qqplot()`, `pca()`, `print.pca()`, `show_correlations()`, `show_weights()`, `spca_screeplot()`, `wachter_qqplot()`

## Examples

```
data(holzinger)
ho_pca = pca(holzinger, screeplot = FALSE)
# the screeplot can be produced from `pca` directly.
# It can be customized by saving it as a `ggplot` object
myscreeplot = scree_plot(ho_pca, return_plot = TRUE)
myscreeplot + ggplot2::geom_point(color = "red")
```

---

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| show_correlations | <i>Show Component Correlations for a pca or spca Object</i> |
|-------------------|-------------------------------------------------------------|

---

## Description

Print and optionally return the mutual correlations among sparse principal components and their correlations with the corresponding principal components.

## Usage

```
show_correlations(
  object,
  type = c("both", "spcs", "pcs"),
  digits = 2,
  print_matrices = TRUE,
  return_matrices = FALSE,
  spca_obj = NULL
)

## S3 method for class 'spca'
show_correlations(
  object,
  type = c("both", "spcs", "pcs"),
  digits = 2,
  print_matrices = TRUE,
  return_matrices = FALSE,
  spca_obj = NULL
)
```

## Arguments

|        |                                                                                                                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | An object of class pca or spca.                                                                                                                                                                                                |
| type   | A character value specifying which correlations to show. Values beginning with "s", "p", or "b" select the mutual sPC correlations, the correlations with the corresponding PCs, or both, respectively. The default is "both". |
| digits | A non-negative integer scalar (default 2). Number of decimal places used for printing. Returned matrices are not rounded.                                                                                                      |

`print_matrices` A logical value (default TRUE). If TRUE, print the requested correlations.

`return_matrices` A logical value (default FALSE). If TRUE, return the requested unrounded numeric matrix or matrices.

`spca_obj` Deprecated alias for `object` (default NULL). Supply only one of `object` and `spca_obj`. Using the old argument name issues a warning.

### Value

If `return_matrices = TRUE`, a numeric matrix when one type of correlation is requested, or a named list of two numeric matrices when `type = "both"`. Otherwise, returns NULL invisibly.

### See Also

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

Other `pca`: [aggregate\\_by\\_group\(\)](#), [change\\_sign\(\)](#), [is.pca\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [print.pca\(\)](#), [scree\\_plot\(\)](#), [show\\_weights\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

### Examples

```
data(holzinger)
ho_cspca = spca(holzinger, n_comps = 3)
show_correlations(ho_cspca)
show_correlations(ho_cspca, type = "s", return_matrices = TRUE)
```

---

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| show_weights | <i>Show the Weights or Contributions of a pca or spca Object</i> |
|--------------|------------------------------------------------------------------|

---

### Description

Show selected nonzero component weights or their unit-L1 contributions.

### Usage

```
show_weights(
  object,
  cols = NULL,
  contribution = TRUE,
  print_list = TRUE,
  return_list = FALSE,
  ...,
  spca_obj = NULL
)
```

```
## S3 method for class 'spca'
show_weights(
  object,
  cols = NULL,
  contribution = TRUE,
  print_list = TRUE,
  return_list = FALSE,
  ...,
  spca_obj = NULL
)

## S3 method for class 'pca'
show_weights(object, ...)
```

### Arguments

|                           |                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>       | An object of class <code>pca</code> or <code>spca</code> .                                                                                                                              |
| <code>cols</code>         | An integer vector selecting components. If a single positive integer is supplied, components <code>1:cols</code> are included. If <code>NULL</code> , all components are included.      |
| <code>contribution</code> | A logical value. If <code>TRUE</code> , show unit-L1 contributions; otherwise, show the original nonzero weights.                                                                       |
| <code>print_list</code>   | A logical value indicating whether to print the result.                                                                                                                                 |
| <code>return_list</code>  | A logical value indicating whether to return the result.                                                                                                                                |
| <code>...</code>          | Additional arguments reserved for S3 method compatibility.                                                                                                                              |
| <code>spca_obj</code>     | Deprecated alias for <code>object</code> (default <code>NULL</code> ). Supply only one of <code>object</code> and <code>spca_obj</code> . Using the old argument name issues a warning. |

### Value

The selected weights or contributions when requested; otherwise `NULL` invisibly.

### Methods (by class)

- `show_weights(pca)`: The weights or contributions of a `pca` object are shown identically those of an `spca` object.

### See Also

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [spca\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

Other `pca`: [aggregate\\_by\\_group\(\)](#), [change\\_sign\(\)](#), [is.pca\(\)](#), [mp\\_qqplot\(\)](#), [pca\(\)](#), [print.pca\(\)](#), [scree\\_plot\(\)](#), [show\\_correlations\(\)](#), [spca\\_screeplot\(\)](#), [wachter\\_qqplot\(\)](#)

spca

*Compute LS-SPCA Components***Description**

Compute least squares sparse principal components (LS-SPCA) from a data matrix or from a covariance/correlation matrix.

**Usage**

```
spca(
  M,
  n_comps = NULL,
  alpha = 0.95,
  ncomp_by_cvexp = NULL,
  method = c("cspca", "uspca", "pspca"),
  var_selection = c("fwd", "bkw", "step"),
  objective = c("cvexp", "r2"),
  intensive = FALSE,
  fat_matrix = NULL,
  fixed_index_list = NULL,
  center_data = FALSE,
  scale_data = FALSE,
  pm_weights = FALSE,
  eps_pm_weights = 1e-04,
  maxiter_pm_weights = 1000,
  pm_varsel = FALSE,
  eps_pm_varsel = 1e-04,
  maxiter_pm_varsel = 500
)
```

**Arguments**

|                |                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M              | A numeric matrix or data frame. If M is square, it is treated as a covariance/correlation matrix and the tall backend is used. Otherwise, M is treated as an $n \times p$ data matrix.                                    |
| n_comps        | A nonnegative integer scalar or NULL (default NULL). Number of components to compute. If NULL, ncomp_by_cvexp is used to determine the number of components. At least one of n_comps and ncomp_by_cvexp must be supplied. |
| alpha          | A numeric scalar in (0, 1] (default 0.95). Target retained proportion used by variable selection.                                                                                                                         |
| ncomp_by_cvexp | A numeric scalar in (0, 1] or NULL (default NULL). If n_comps = NULL, components are computed until cumulative variance explained reaches this value.                                                                     |
| method         | A character vector (default first element "cspca"). LS-SPCA variant. Accepted values are "cspca", "uspca", and "pspca"; only the first letter is used.                                                                    |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>var_selection</code>      | A character vector (default first element "fwd"). Variable-selection algorithm. Values starting with "f" use forward selection, values starting with "b" use backward elimination, and values starting with "s" use forward-stepwise selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>objective</code>          | A character vector (default "cvexp" for all components). Stopping criterion for variable selection. Values starting with "r" use the squared-correlation criterion; values starting with "c" use cumulative variance explained.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>intensive</code>          | A logical value (default FALSE). If TRUE, the tall backend uses intensive forward CVEXP selection. This option is not available for fat matrices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>fat_matrix</code>         | A logical value or NULL (default NULL). If NULL, data matrices with more columns than rows use the fat backend, and all other inputs use the tall backend. If TRUE, the fat backend is requested. If FALSE, the tall backend is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>fixed_index_list</code>   | A list of integer-valued vectors, a factor, or NULL (default NULL). Each list element gives the fixed variable indices for the corresponding component. A NULL or empty element requests variable selection for that component. A list of length one fixes only the first component; to fix a later component, include NULL or empty placeholders for the preceding components. Index sets may overlap and need not include every variable. Indices are 1-based. A factor remains available as a shorthand for assigning variables to component-specific groups. Duplicate indices within a component are removed with a warning. If the list is longer than the number of components, extra elements are ignored with a warning. |
| <code>center_data</code>        | A logical value (default FALSE). If TRUE, center data-matrix columns before fitting. Ignored when M is treated as a covariance/correlation matrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>scale_data</code>         | A logical value (default FALSE). If TRUE, scale data-matrix columns before fitting. Ignored when M is treated as a covariance/correlation matrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>pm_weights</code>         | A logical value (default FALSE). If TRUE, use the power method for PC and sparse-weight eigenvectors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>eps_pm_weights</code>     | A positive numeric scalar (default 1e-4). Convergence tolerance for <code>pm_weights</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>maxiter_pm_weights</code> | A positive integer scalar (default 1000). Maximum number of iterations for <code>pm_weights</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>pm_varsel</code>          | A logical value (default FALSE). If TRUE, use the power method inside variable selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>eps_pm_varsel</code>      | A positive numeric scalar (default 1e-4). Convergence tolerance for <code>pm_varsel</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>maxiter_pm_varsel</code>  | A positive integer scalar (default 500). Maximum number of iterations for <code>pm_varsel</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Details

Data matrices are routed to the tall or fat C++ backend. Square matrices are treated as covariance/correlation matrices and use the tall backend.

Variable selection is controlled by `var_selection`, `objective`, and `intensive`.

| var_selection                   | objective | Algorithm                                                    |
|---------------------------------|-----------|--------------------------------------------------------------|
| "fwd"                           | "r2"      | Forward selection with squared-correlation stopping          |
| "bkw"                           | "r2"      | Backward elimination with squared-correlation stopping       |
| "step"                          | "r2"      | Forward-stepwise selection with squared-correlation stopping |
| "fwd"                           | "cvexp"   | Forward selection with CVEXP stopping                        |
| "bkw"                           | "cvexp"   | Backward elimination with CVEXP stopping                     |
| "step"                          | "cvexp"   | Forward-stepwise selection with CVEXP stopping               |
| intensive = TRUE requires "fwd" | "cvexp"   | Intensive forward CVEXP selection                            |

The fat backend currently supports regression-based forward variable selection only: `var_selection = "f"` and `intensive = FALSE`. Other combinations generate an error.

If both `n_comps` and `ncomp_by_cvexp` are supplied, `n_comps` takes precedence and automatic stopping by cumulative variance explained is disabled.

The returned object is documented in [spca\\_object](#).

## Value

An object of class `spca`.

## See Also

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\\_object](#), [summary.pca\(\)](#)

## Examples

```
data(holzinger)
#default
ho_cspca = spca(holzinger, n_comps = 4)
#uncorrelated components and subsets determined using CVEXP as stopping rule
ho_uspca = spca(holzinger, n_comps = 4, method = "uspca",
               objective = "cvexp")
#fix indices for components 1 and 3 and select them for component 2
ho_fixed = spca(holzinger, n_comps = 3,
               fixed_index_list = list(c(1, 2), NULL, c(2, 4, 6)))
```

---

spca\_object

*Sparse Principal Component Analysis Object*

---

## Description

Objects of class `spca` are returned by the fitting functions [spca\(\)](#), [pca\(\)](#), and [new\\_spca\(\)](#). Objects returned by [pca\(\)](#) also inherit from class `spca_pca`.

## Components

An object of class `spca` is a list with the following elements:

**weights**  $p \times r$  matrix of sparse weights.

**contributions**  $p \times r$  matrix of weights scaled to unit  $L_1$  norm within each sPC.

**n\_comps** Number of sPCs.

**cardinality** Number of nonzero weights in each sPC.

**vexp** Variance explained by each sPC.

**vexp\_pc** Variance explained by the corresponding PCs.

**cvexp** Cumulative variance explained by the sPCs.

**rvexp** Ratio of `vexp` to the variance explained by the corresponding PC.

**rcvexp** Ratio of `cvexp` to the cumulative variance explained by the corresponding PCs.

**cor\_with\_pc** Correlation between each sPC and the corresponding PC.

**tot\_var** Total variance of the data.

**weights\_list** List of nonzero weight vectors, one per sPC.

**spc\_cor**  $n_{comps} \times n_{comps}$  correlation matrix of the sPC scores.

**indices** List of variable indices with nonzero weights, one per sPC.

**scores** Optional matrix of sPC scores, returned only when a data matrix is supplied.

**parameters** List of parameters used to compute an `spca()` fit.

**call** Matched call used to compute an `spca()` fit.

**eigenvalues** For `pca()` objects, the available PCA eigenvalues.

**n\_obs** For `pca()` objects, the number of observations when available.

**method\_name** For `new_spca()` objects, an optional method label.

For backward compatibility, methods also accept objects from earlier package versions containing `loadings` and `loadings_list` instead of `weights` and `weights_list`.

## See Also

Other `spca`: `aggregate_by_group()`, `change_loadings_sign_spca()`, `change_sign()`, `change_weights_sign_spca()`, `compare_spca()`, `is.spca()`, `new_spca()`, `plot.pca()`, `print.pca()`, `show_contributions_spca()`, `show_correlations()`, `show_weights()`, `spca()`, `summary.pca()`

---

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| summary.pca | <i>Summarize the Variance Explained and Sparsity of a pca or spca Object</i> |
|-------------|------------------------------------------------------------------------------|

---

## Description

Print and, optionally, return summary statistics for evaluating an spca object and comparing it with the corresponding PCA solutions. It can also print the summaries of a pca object.

## Usage

```
## S3 method for class 'pca'
summary(object, ...)

## S3 method for class 'spca'
summary(
  object,
  cols = NULL,
  contributions = TRUE,
  variance_metrics = c("both", "cumulative_relative", "relative", "none"),
  min_weight = FALSE,
  cor_with_pc = TRUE,
  return_table = FALSE,
  print_table = TRUE,
  thresh_card = 1e-08,
  ...
)
```

## Arguments

|                  |                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object           | An object of class spca or pca.                                                                                                                                                      |
| ...              | Further arguments. These are currently unused and trigger an error if supplied.                                                                                                      |
| cols             | An integer vector of component indices or NULL. If NULL (the default), all available components are included. If a single integer is supplied, components 1:cols are included.       |
| contributions    | A logical value (default TRUE). If TRUE, minimum nonzero values are computed from percentage contributions; otherwise, they are computed from weights.                               |
| variance_metrics | A character vector (default first element "both"). Controls which relative variance metrics are included. Accepted values are "relative", "cumulative_relative", "both", and "none". |
| min_weight       | A logical value (default FALSE). If TRUE, include the minimum nonzero weight or contribution.                                                                                        |
| cor_with_pc      | A logical value (default FALSE). If TRUE, include correlations between sPCs and the corresponding PCs when available.                                                                |

|              |                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| return_table | A logical value (default FALSE). If TRUE, return the raw numeric summary matrix.                                                           |
| print_table  | A logical value (default TRUE). If TRUE, print the formatted summary table.                                                                |
| thresh_card  | A numeric scalar (default 1e-8). Values with absolute magnitude at or below this threshold are treated as zero when computing cardinality. |

## Details

For each component, the following summaries can be computed:

|                     |                                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Vexp                | The percentage variance explained.                                                |
| Cvexp               | The percentage cumulative variance explained.                                     |
| Rvexp               | The variance explained relative to the corresponding PC.                          |
| Rcvexp              | The cumulative variance explained relative to the corresponding PCs.              |
| Card                | The cardinality, that is the number of non zero weights.                          |
| Min weight/Min cont | The minimum absolute value of the nonzero weights or contributions, if requested. |
| r                   | The correlation between sPCs and the corresponding PCs, if requested.             |

## Value

If return\_table = TRUE, returns a numeric matrix with the selected summary statistics. Otherwise, returns NULL invisibly.

## Functions

- `summary(pca)`: Summarize a `pca` object using the shared summary method.

## See Also

Examples in [aggregate\\_by\\_group](#).

Other `spca`: [aggregate\\_by\\_group\(\)](#), [change\\_loadings\\_sign\\_spca\(\)](#), [change\\_sign\(\)](#), [change\\_weights\\_sign\\_spca\(\)](#), [compare\\_spca\(\)](#), [is.spca\(\)](#), [new\\_spca\(\)](#), [plot.pca\(\)](#), [print.pca\(\)](#), [show\\_contributions\\_spca\(\)](#), [show\\_correlations\(\)](#), [show\\_weights\(\)](#), [spca\(\)](#), [spca\\_object](#)

## Examples

```
data(holzinger)
ho_cspca = spca(holzinger, n_comps = 2)
summary(ho_cspca)
```

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